

InterLens Web UI Specification & Design Blueprint

InterLens features a modern, high-density, dark-mode visual interface styled like a professional IDE / mechanistic interpretability workbench.

1. Visual Design Tokens & Palette

Token	Hex Value	Application
--bg-darkest	#0A0D14	Dashboard main canvas background
--bg-card	#121824	Panels, drawers, and card containers
--bg-hover	#1E2638	Hover state for lists and buttons
--border-color	#2A344A	Sleek border lines and panel splitters
--text-primary	#F1F5F9	Main headers and primary text
--text-muted	#94A3B8	Subtitles, labels, and secondary metadata
--accent-cyan	#38BDF8	Primary active tab & Logit Lens highlights
--accent-purple	#A855F7	Attention & Residual Stream highlights
--accent-green	#34D399	Neuron activation & positive patch recovery
--accent-rose	#FB7185	Negative patch recovery & corrupted state

2. Layout Structure

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| [Logo] InterpLens | Model: [ Llama-3.2-1B v ] | Prompt: [ The Eiffel Tower... ] [Run] |
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| SIDEBAR | MAIN PANEL (Persistent Token Strip at Top) | CONTEXT CONTROL | | |
|         | [ <|EOS|> ] [ The ] [ Eiffel ] [ Tower ] [ is ] | PANEL |
| Overview |-----| | | |
| Logit Lens | | | |
| Residual | [ INTERACTIVE CHART VIEW ] | Layer: [ slider ] |
| Attention | (Ribbon / Heatmap / Grid / Arc) | Head: [ slider ] |
| Neurons | | | |
| Patching | | | |
| Causal Tr. | | | |
| SAE Feat. | | | |
| Compare | | | |
+-----+
| BOTTOM | [Raw Tensor Data Inspector] [Session History] | GPU Memory: 1.2GB |
+-----+
```

3. Chart Specifications per Feature

3.1 Overview / Token Attribution

- **View:** Prompt rendered as tokens.
- **Visual:** Text tokens highlighted with variable opacity background matching attribution score.
- **Controls:** Attribution method dropdown (Gradient, Attention Rollout).

3.2 Logit Lens

- **View 1 (Ribbon):** X-axis = Layer (0..L), Y-axis = Confidence / Rank. Curves represent top candidate tokens evolving across layers.
- **View 2 (Heatmap Matrix):** Y-axis = Layer (0..L), X-axis = Token Position (0..T). Cell text = Top-1 predicted token at that layer; Cell color = Probability magnitude.

3.3 Residual Stream

- **Heatmap:** X-axis = Token Position, Y-axis = Layer. Color gradient = L2 norm / variance of residual stream vector.
- **PCA Trajectory:** 2D scatter plot showing vector trajectory from Layer 0 to Layer \$L\$ for a clicked token.

3.4 Attention Heads

- **Grid View:** Small N×N thumbnail heatmaps for all heads in selected layer.
- **Detail View:** Full-size annotated matrix with token labels on both axes.

- **Arc View:** Prompt text laid out horizontally; curved arcs connecting query and key tokens, with arc thickness scaled to attention weight.

3.5 Neuron Activation Panel

- **Bar Chart:** Ranked horizontal bars showing top-firing neurons for selected token.
- **Token Lighting Strip:** Text display where each token is shaded by how strongly the selected single neuron fired.

3.6 Activation Patching & Causal Tracing

- **Patch Heatmap:** X-axis = Token Position, Y-axis = Layer. Color scale = Logit difference recovered (Diverging Red-Blue).
- **Causal Trace Sweep:** Smoothed gradient heatmap representing automated ROME sweep importance.

3.7 Model Comparison

- **Split Screen:** Two identical charts side-by-side with synchronized controls (layer sliders, zoom/pan).
- **Diff View:** Single heatmap showing tensor differences ($\text{Model_A} - \text{Model_B}$).